MARK SCHEME · TEACHER COPY

11BR · Topic 4: Processor Fundamentals

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · May/June 2024 Paper 13, original Q3 (a),(b) [7 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(a)</td><td>1 mark for each correct answer:
<table>
<tr><th>Program Number</th><th>Code</th><th>ACC Content</th></tr>
<tr><td>1</td><td>LDM #50
INC ACC
SUB #1</td><td>50</td></tr>
<tr><td>2</td><td>LDI 51
ADD 52</td><td>97</td></tr>
<tr><td>3</td><td>LDR #2
LDX 50
DEC ACC</td><td>48</td></tr>
<tr><td>4</td><td>LDD 52
SUB 54
INC ACC</td><td>44</td></tr>
</table></td><td>4</td></tr>
<tr><td>1(b)</td><td>1 mark for each correct answer:
<table>
<tr><th>Instruction Number</th><th>Instruction</th><th>ACC Content</th></tr>
<tr><td>1</td><td>LSL #2</td><td>1111 1100</td></tr>
<tr><td>2</td><td>XOR 100</td><td>1111 0010</td></tr>
</table></td><td>3</td></tr>
</table>

3	AND 103	**0011 0111**

Question 2 · May/June 2025 Paper 12, original Q1 (a),(b),(c) [9 marks]

Question	Answer	Marks
2(a)	**1 mark** for each correct row	**3**

Incorrect statement number	Corrected statement
1	The Program Counter (PC) stores the **address of** the next instruction to be fetched from memory.
3	The Control Unit (CU) sends signals to other components on the **control** bus.
4	The Memory Data Register (MDR) **holds data to be stored in** the memory address in the MAR. // The Memory Data Register (MDR) **holds data read from** the memory address in the MAR

2(b)	**1 mark** for name, **1 mark** for corresponding role e.g. • Current Instruction Register // CIR • To store the instruction to be decoded / executed next • Status register // SR • To contain bits that can be referenced individually to indicate a state or event • Interrupt register • To store details of any interrupts that have occurred	**2**
2(c)	**1 mark** each to **max 4** • At the start / end of FE cycle the interrupt register is checked • The priority of any interrupts waiting is checked • if the priority of the interrupt is higher than the current process • the contents of the registers are stored on the stack • The relevant Interrupt Service Routine (ISR) / interrupt handler is called to process the interrupt • When the ISR has finished, a further check is made for higher priority interrupts • if no more interrupts of higher priority, the register contents are restored and the next FE cycle continues	**4**

Question 3 · May/June 2025 Paper 12, original Q7 (a),(b) [8 marks]

Question	Answer	Marks
3(a)(i)	**1 mark** for each shaded section	**3**

Instruction address	ACC	Memory address				
		10	11	12	13	14
		12	11	10	22	22
100	12					
101	22					
102			22			
103						
104						
107		22				

3(a)(ii)	**1 mark** from: • The number 10 will be loaded into the ACC instead of 12 / contents of address 10 • The addition will give 20 not 22 so the comparison will fail and result in an infinite loop	**1**

3(a)(iii)	**1 mark** for name of mode, **1 mark** for matching description e.g. • Indirect • The operand points to the memory location that contains the address of the data • Indexed • The address of the data is formed by adding the contents of the Index Register (IX) to the operand • Relative • The address is calculated using its distance from a base address	**2**
3(b)(i)	**1 mark** for: 1111 1111	**1**
3(b)(ii)	**1 mark** for: 0000 0011	

Question 4 · May/June 2025 Paper 13, original Q5 (a),(b) [9 marks]

Question	Answer	Marks
4(a)(i)	**1 mark** for each shaded section	**3**

Instruction address	ACC	Memory address			
		50	51	52	53

		47	48	49	50
500	50				
501	49				
502					
504					
501	48				
502					
503					
505		48			
506	38				
507			38		

4(a)(ii)	**1 mark** for addressing mode, **1 mark** for matching description • Direct • The operand is the address of the data • Indirect • The operand points to the memory location which is the address of the data • Indexed • The address of the data is formed by adding the contents of the Index Register (IX) to the operand	**4**
4(b)	**1 mark** for each correct instruction e.g. AND B00000000 / #0 / &0 OR B10000000 / #128 / &80	**2**

Question 5 · Oct/Nov 2023 Paper 12, original Q9 (a),(b) [8 marks]

Question	Answer	Marks
5(a)	**1 mark** for each bullet point • `IN` - Input **and** output of data • `ADD` - Arithmetic operations • `JPE` - Unconditional and conditional instructions • `CMI` - Compare instructions	**4**

5(b)

1 mark for each set of shaded rows

4

Instruction address	ACC	IX	Memory address				Output
			100	101	110	111	
			0	0	51	65	
10		0					
11	49						
12				49			
13	51						
14							
15							
16	49						
17	65						
18		1					
19			65				
20							
13	65						
14							
15							
21							A
22							

Question 6 · Oct/Nov 2023 Paper 13, original Q9 (a),(b),(c),(d) [13 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>6(a)(i)</td><td>1 mark for each bullet point (max 3).

• Control unit (CU)
• Arithmetic and Logic Unit (ALU)
• Immediate Access Store (IAS)
• (System clock)</td><td>3</td></tr>
<tr><td>6(a)(ii)</td><td>1 mark for each bullet point (max 2).

• Special purpose registers hold the status of a program whereas general purpose registers hold the temporary data while performing operations.
• Special purpose registers are specialised for a specific use, whereas general purpose registers are used for any purpose.
• General purpose registers can be used by most instructions, whereas special purpose can only be used by certain instruction</td><td>2</td></tr>
<tr><td>6(b)</td><td>1 mark for each correct answer (shaded cells)

<table><tr><th>Step</th><th>Description</th></tr><tr><td>PC ← [PC] + 1</td><td>The address in PC is incremented.</td></tr><tr><td>MDR ← [[MAR]]</td><td>The data in the location pointed to by the MAR is copied to the MDR.</td></tr></table></td><td>4</td></tr>
</table>

		MAR ← [PC]	The **contents** of PC are copied to the MAR.
		CIR ← [MDR]	The contents of MDR are copied into CIR.
6(c)	**1 mark** for an appropriate example. Examples: • Division by zero • Runtime error • Out of memory bounds • Program requesting an external device / input • Buffer overflow		**1**
6(d)	**1 mark** for each letter in the correct position (2, 4 and 6) **D** **A** **B**		**3**

Question 7 · Oct/Nov 2025 Paper 12, original Q3 (a),(b),(c) [10 marks]

Question	Answer	Marks
7(a)(i)	**1 mark** • The value of the operand is an offset value which is added to another base value to give the address from which the contents are loaded to the accumulator	**1**

7(a)(ii)	**1 mark** per bullet point, **max 2** marks • Program counter (PC) • Memory Data Register (MDR) • Memory Address Register (MAR) • Current Instruction Register (CIR) • Status Register	**2**
7(b)	**1 mark** for each correct value, **max 3** marks **Instruction** / **Value in ACC** `LDM #98` / 98 `LDI 101` / 8 `LDX 100` / 32	**3**
7(c)	**1 mark** per bullet point, **max 2** marks for each Clock Speed: • Processor can perform more F-E cycles per second • ... so more instructions / data can be processed each second Cache Memory: • Can store more of the most frequently used instructions • … which reduces the need to access slower RAM	**4**

Question 8 · Oct/Nov 2025 Paper 13, original Q6 (a),(b),(c) [11 marks]

Question	Answer	Marks

8(a) — **1 mark** for each correct row — **4**

Current contents of the ACC	Instruction	Contents of the ACC after the execution of the instruction
0000 1111	AND 101	0000 1110
0000 0000	LDM #100	0110 0100
0000 0001	XOR &F1	1111 0000
0001 0001	CMP 101	0001 0001

8(b) — **1 mark** for each bullet point, **max 4** marks — **4**

- MAR stores the address of the next instruction / data **to be read from / or written to memory**
- The address is received from the Program Counter (PC)
- The MDR stores the data / instruction in the address stored in the MAR **which has been read / written**
- The instruction passes to the CIR for decoding and executing

8(c) — **1 mark** for each bullet point, **max 3** marks — **3**

- Data movement
- Arithmetic operations
- Conditional and unconditional (jump) instructions